

ECI in sub-Saharan Africa: Distribution, Change, and Key Drivers



SUMMARY

This report looks at the **Economic Complexity Index (ECI)** for a group of African countries over the period **1995 to 2018**. ECI is used to show how complex and developed an economy is based on the type of productive activities and knowledge present in that country.

The purpose of this report is to understand how ECI is distributed across countries, how it has changed over time, and which variables seem to be linked to it. The analysis also looks at the geographical spread of ECI across the sample.

Before the analysis, the data was cleaned by replacing missing values with the median for each variable. This was done to keep the dataset complete while reducing the effect of extreme values.

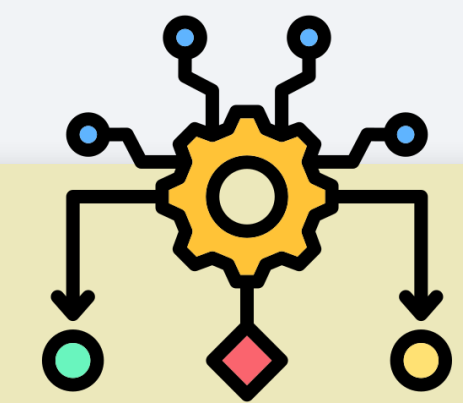
After cleaning, the final dataset contained 500 observations, covering 30 countries over 24 years.



OBJECTIVE

- To examine how ECI is distributed across the countries in the dataset
- To analyse how ECI changed over time
- To explore which variables are related to ECI
- To show the geographical distribution of ECI across the sample

VARIABLES



1. Country Name
2. Time
3. GDP (constant 2010 US\$)
4. GDP per capita (constant 2010 US\$)
5. Energy use (kg of oil equivalent) per \$1,000 GDP
6. CO2 emissions (kt)
7. CO2 emissions (metric tons per capita)
8. International tourism, number of arrivals
9. Air transport, passengers carried
10. Coal rents (% of GDP)
11. ECI

DATA OVERVIEW

30 Number of Countries	500 Number of Observations	11 Number of Variables	-0.7 Average ECI (Latest Year)	Tanzania Best Improvement Country	Zimbabwe Worst Declining Country
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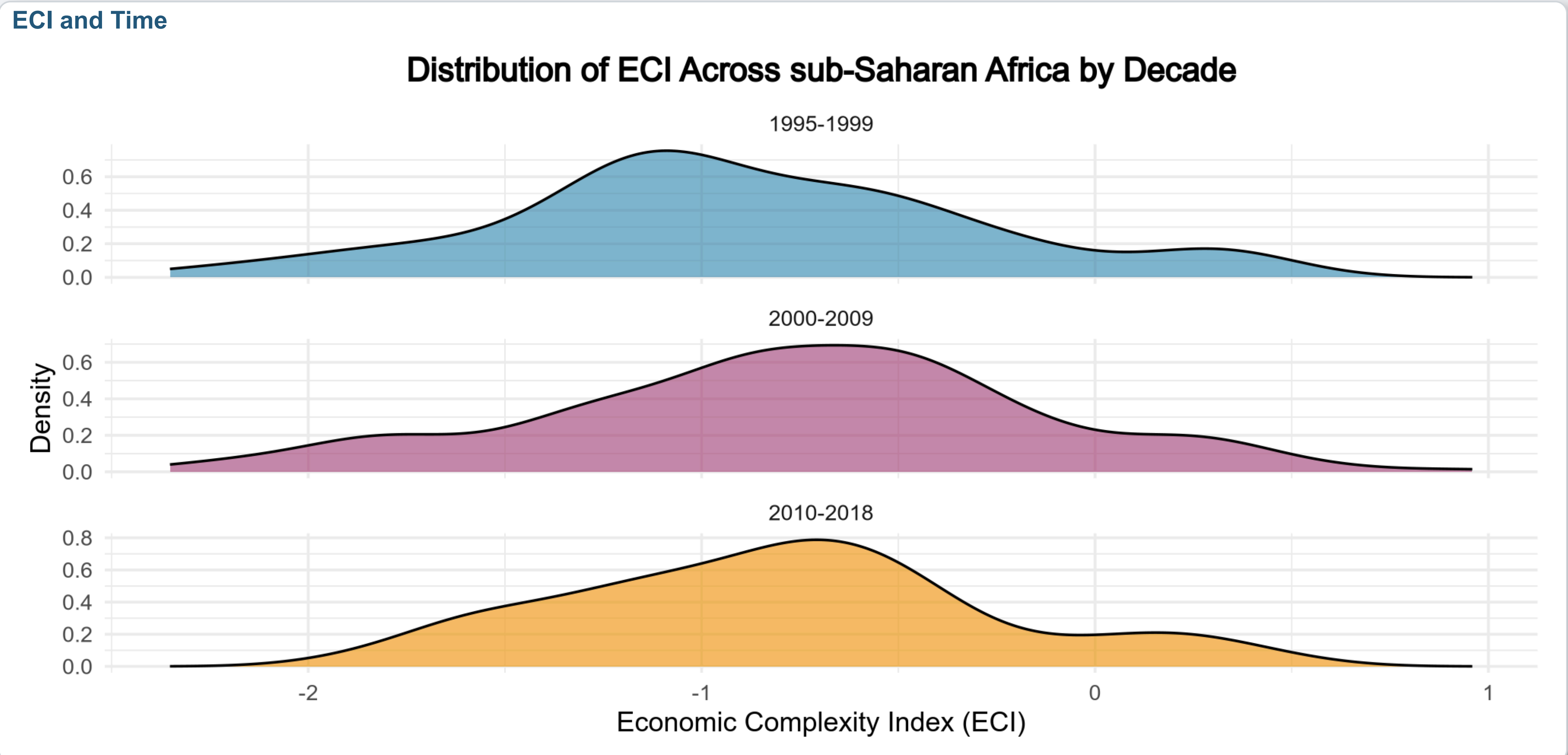
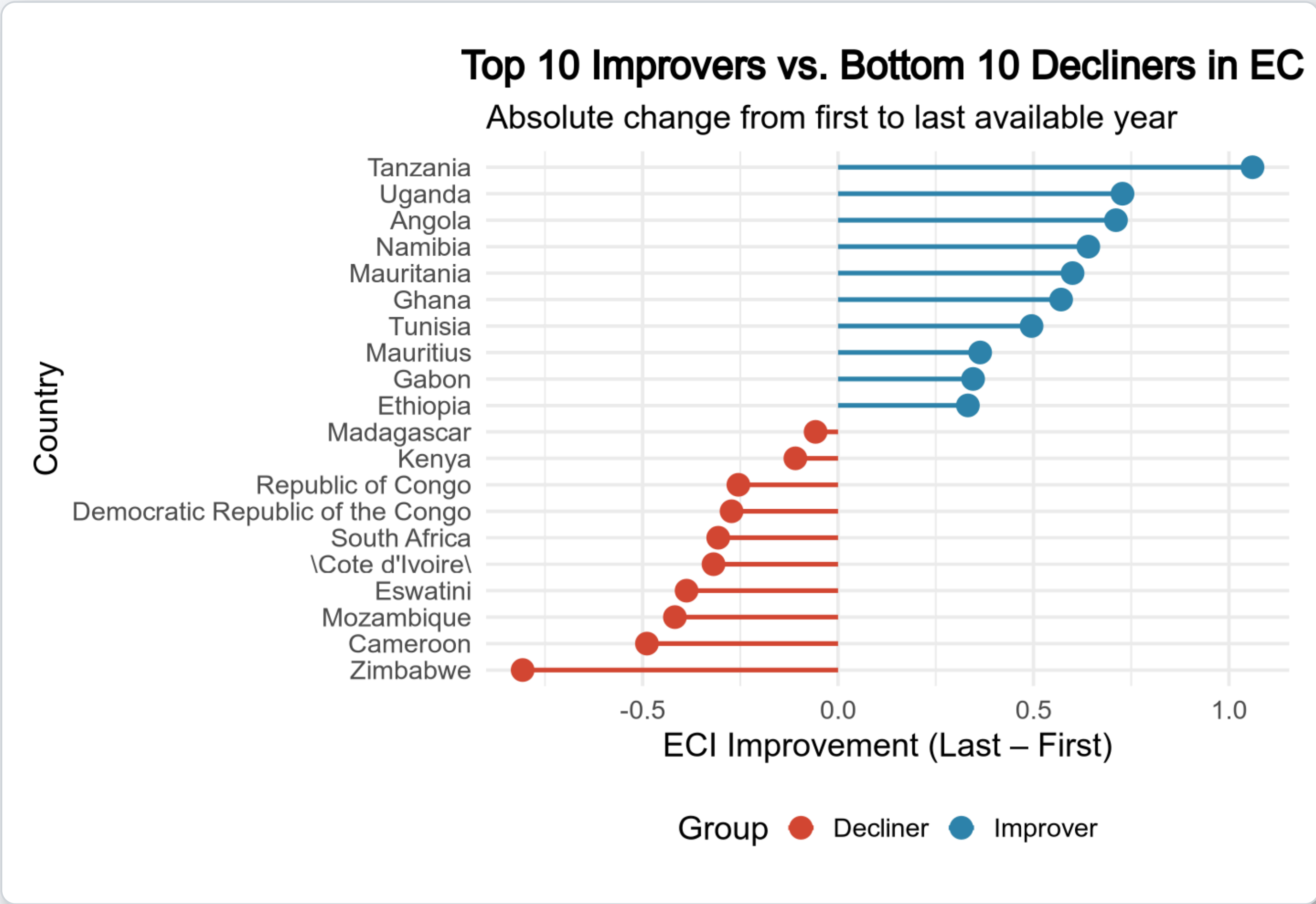
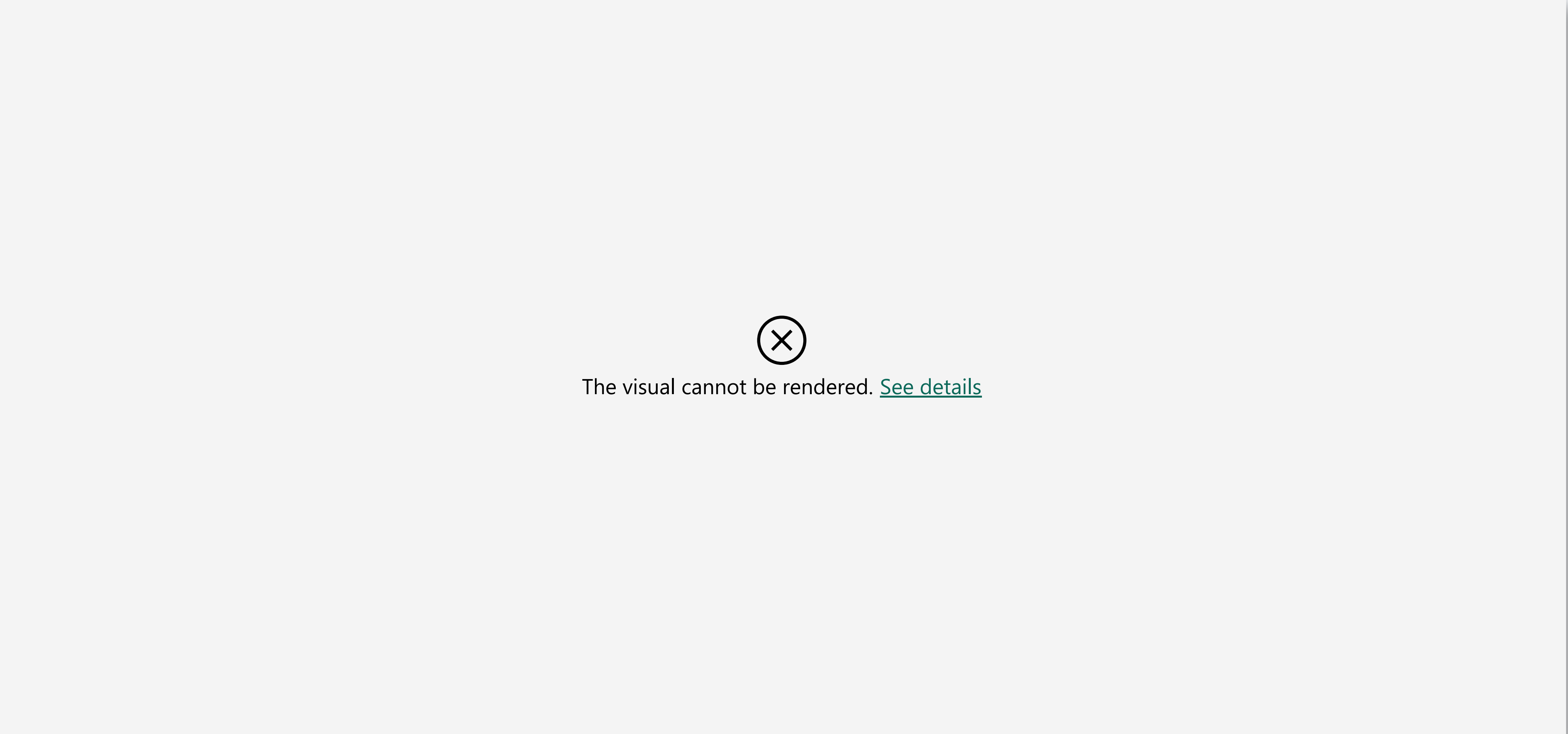
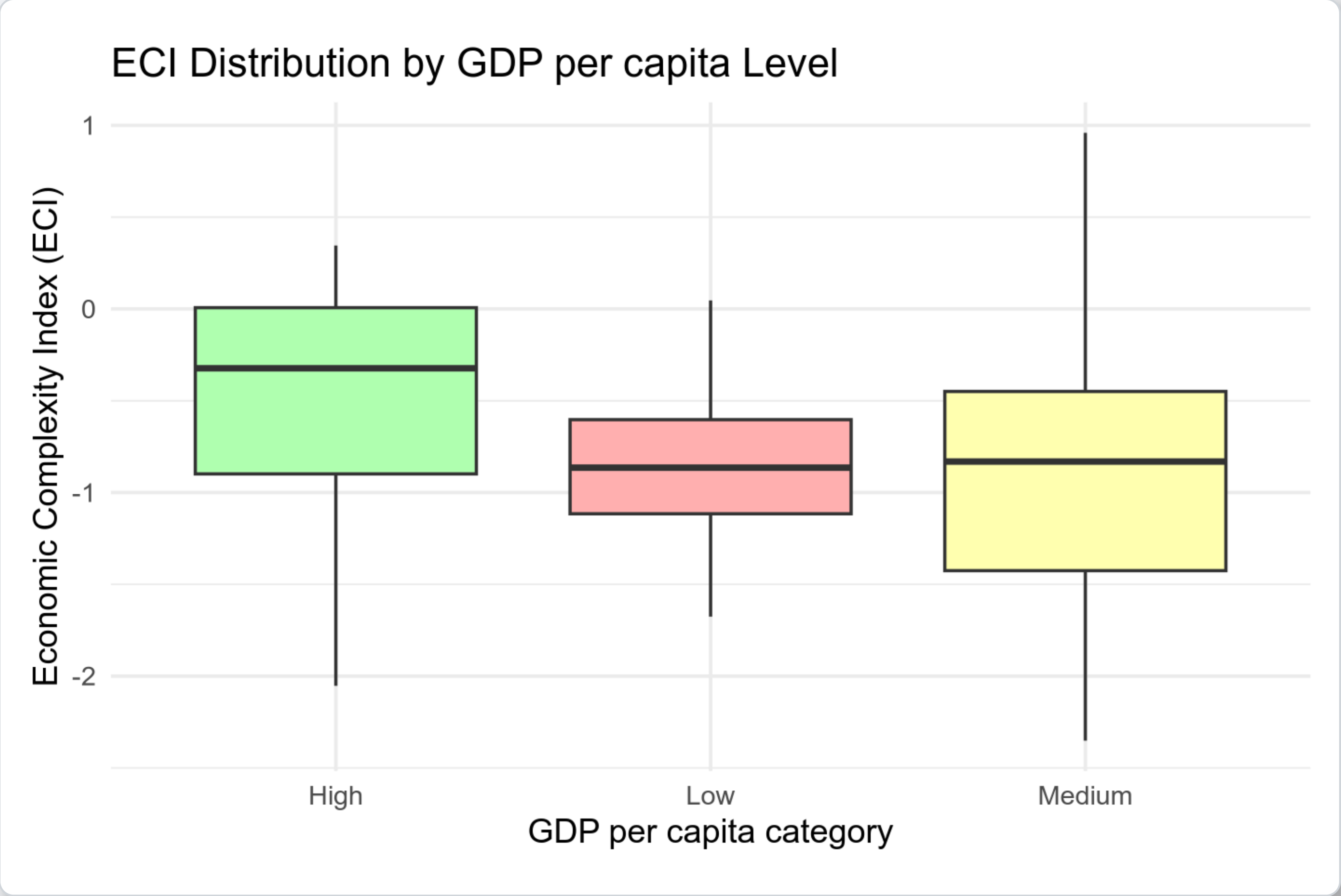
Dataset Structure:

The dataset is structured as panel data (country-year). A hierarchical imputation approach was used: country-level median first, followed by year-level median, and finally global median as fallback.

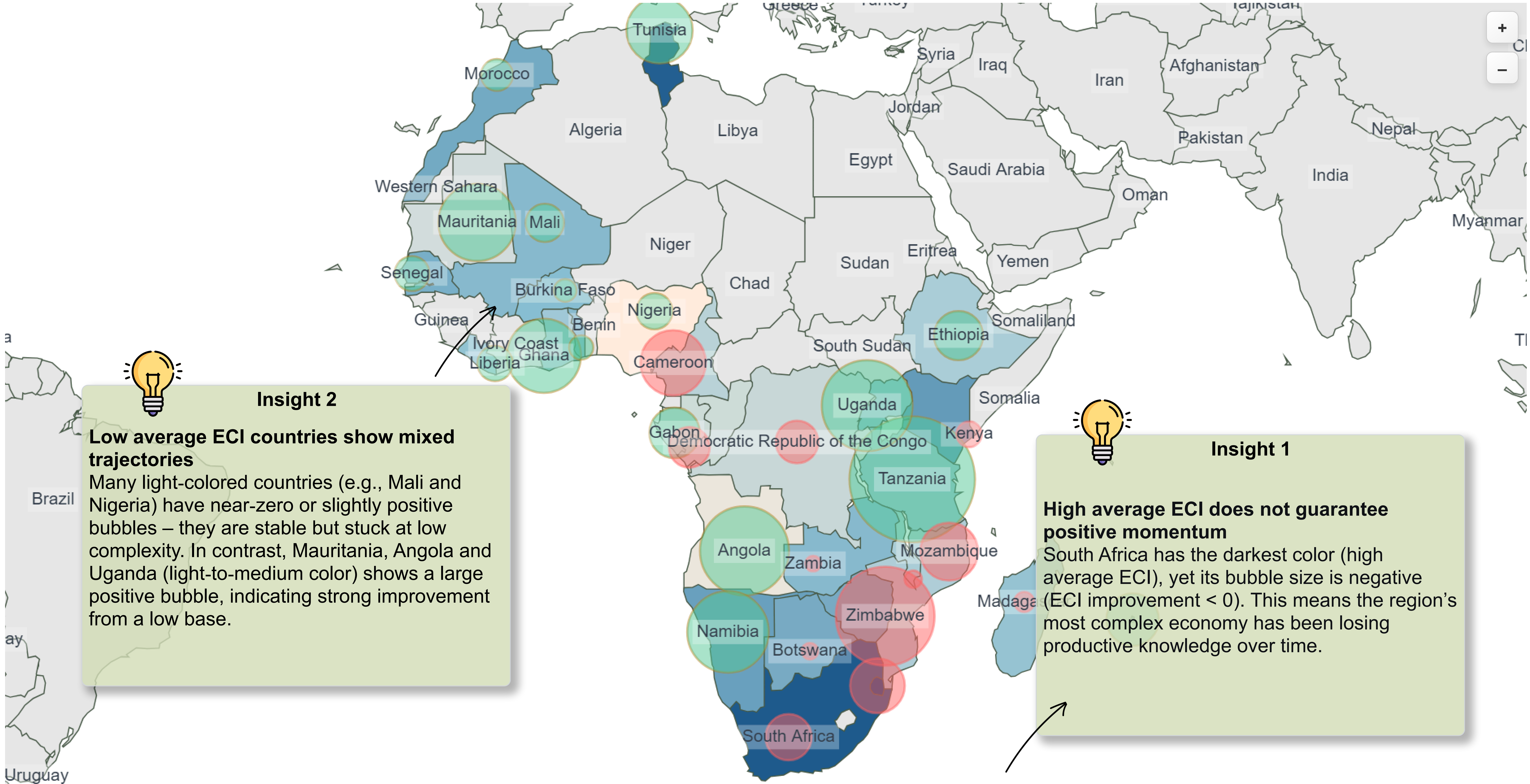
Missing Value Imputation:

Median imputation was chosen because it is more robust to outliers than the mean, particularly in skewed economic data. Indicators such as GDP and carbon emissions often have heavy-tailed distributions, where extreme values can distort the average. Using the median therefore provides a more reliable representation of central tendency and helps preserve the overall structure of the data.

Imputation Method			
Variable	Imputation Method	Split Level	Justification
Air Transport	Median	Year → Country → Global	Global shocks dominate; year-level imputation preferred.
CO2 Emissions (kt)	Median	Country → Year → Global	Large emitters create skew; median avoids distortion.
CO2 per capita	Median	Country → Year → Global	Reflects structural differences; median robust to outliers.
Coal Rents	Zero	N/A	Absence of coal sector implies true zero.
ECI	Median	Country → Year → Global	Slow-moving index; median maintains stability.
Energy Use	Median	Country → Year → Global	Efficiency differs structurally; median is robust.
GDP	Median	Country → Year → Global	Highly skewed across countries; median reduces influence of extreme economies.
GDP per capita	Median	Country → Year → Global	Country-specific income structures; median preserves distribution.
Tourism	Median	Year → Country → Global	Affected by global shocks; year median captures systemic effects.



Average of ECI and ECI Improvement (Bubbles) by Country



Do richer, cleaner, or more visited countries have higher ECI?

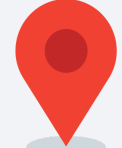
● High Development & High ECI ● Developing Economies ● Low Development but High ECI ● High Development & Low ECI





Year

All

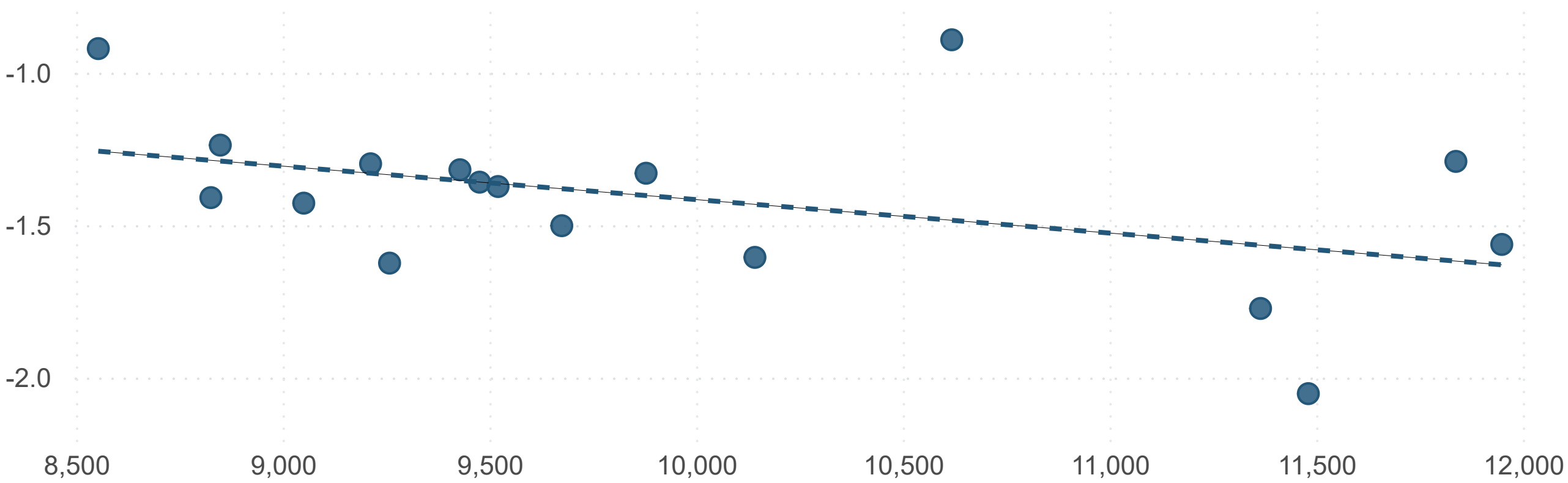


Category, Country Name

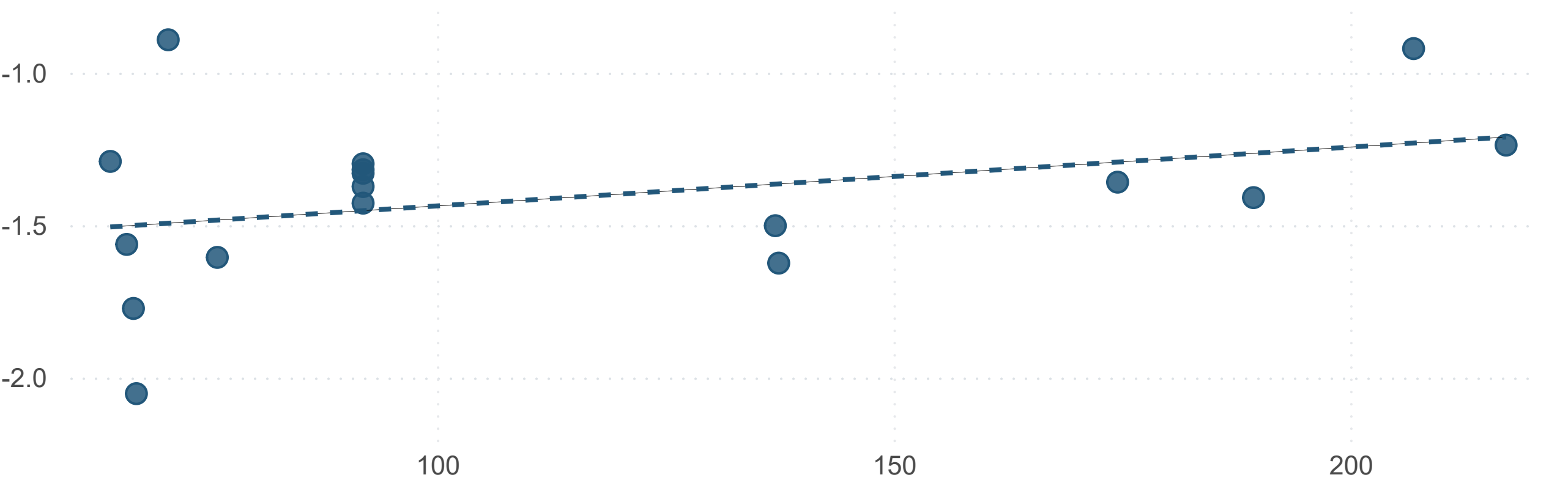
High Development & Low ECI



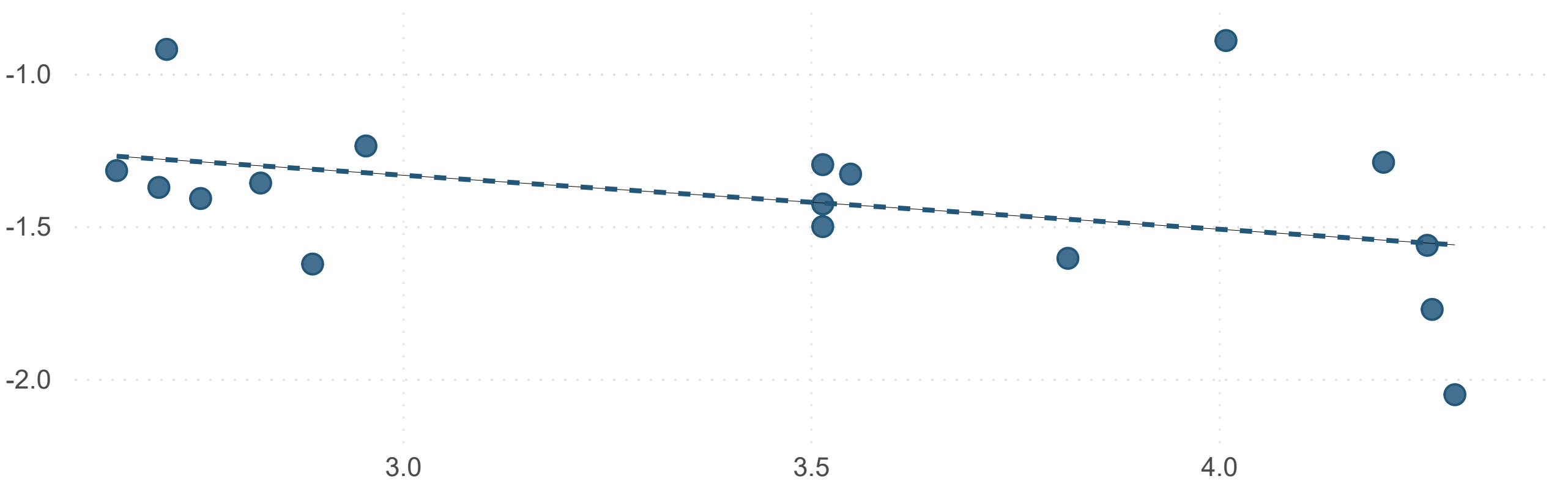
ECI vs GDP per capita



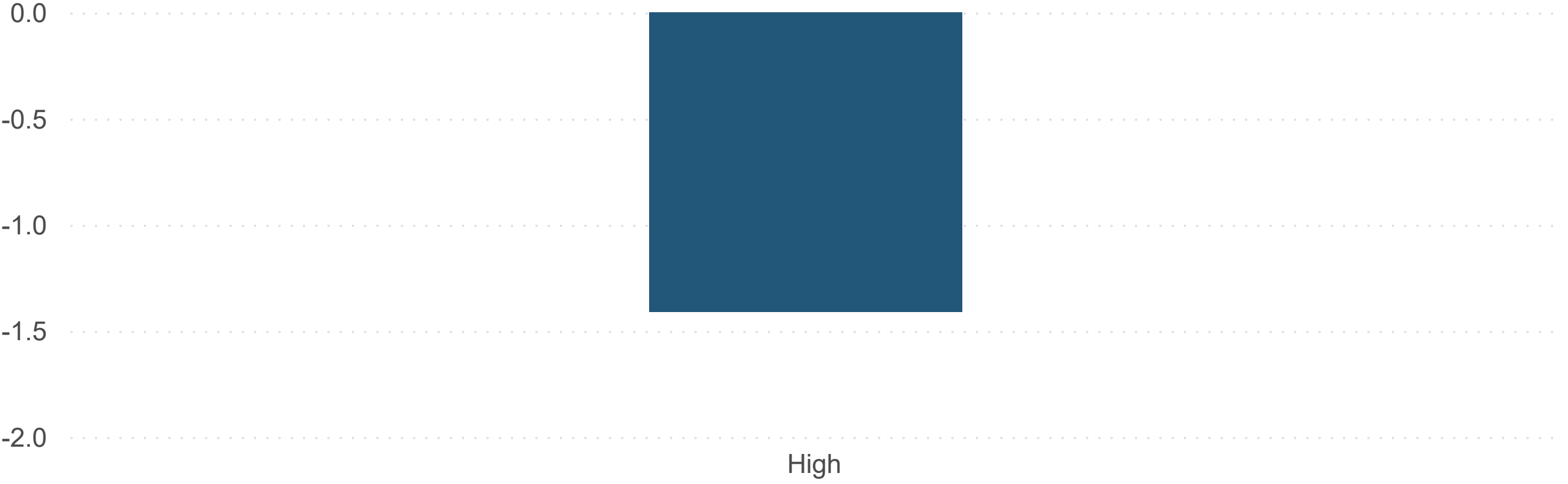
ECI vs Energy use



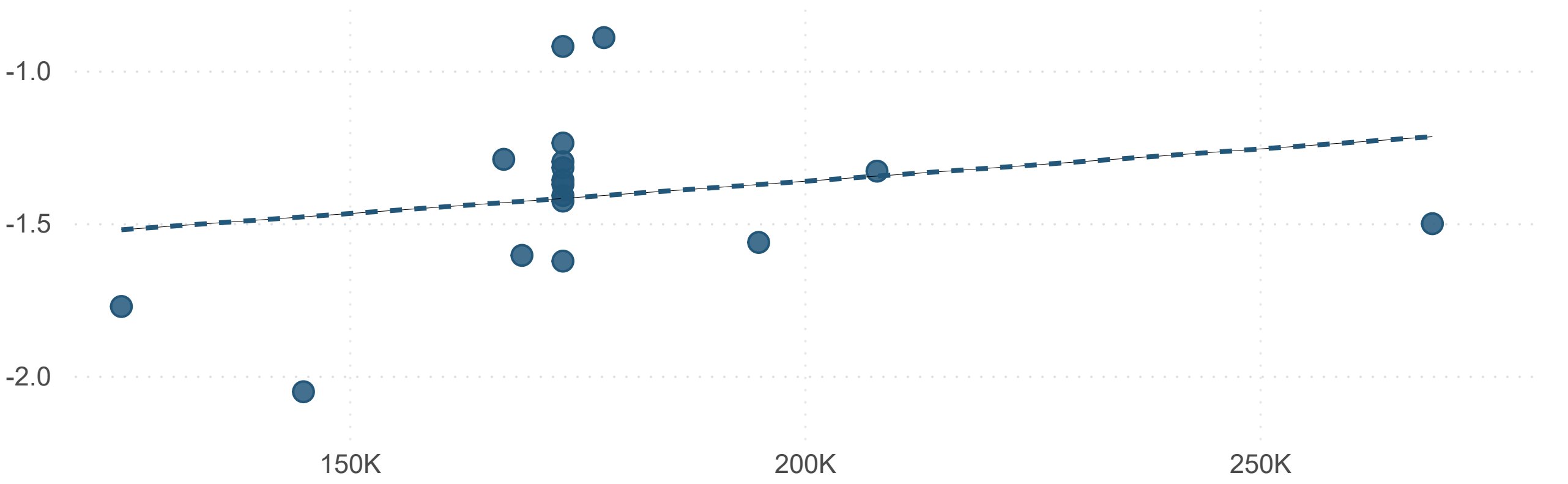
ECI vs CO2 emissions (metric tons per capita)



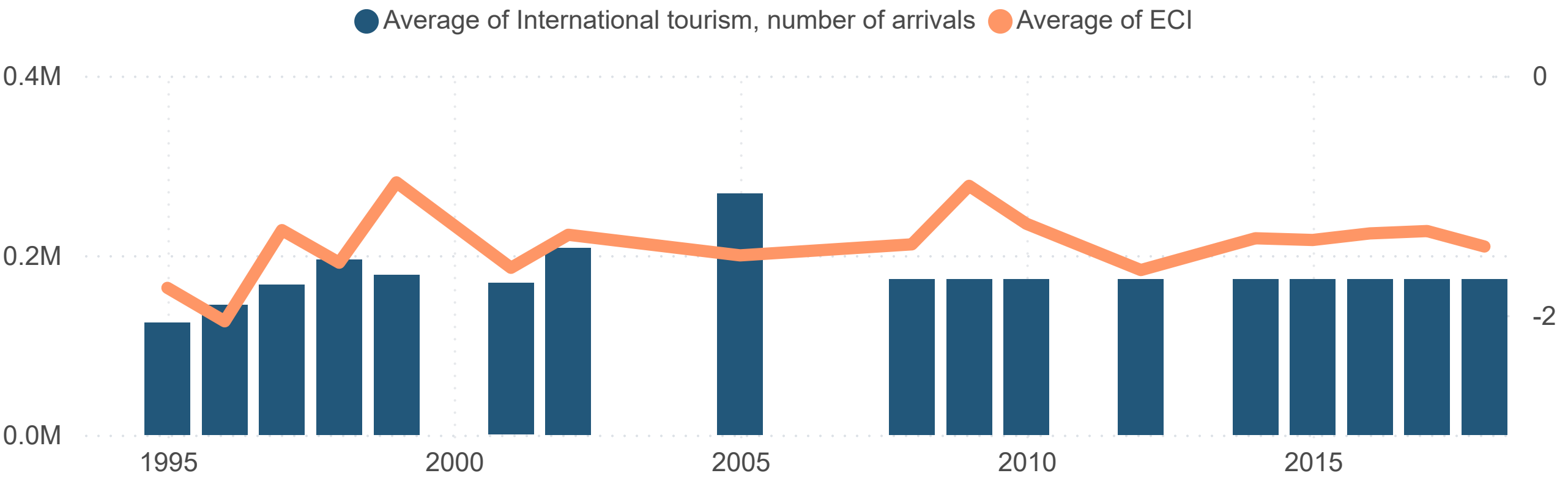
Average of ECI by GDP Category



ECI vs International tourism



ECI and Tourism Arrivals Over Time



Conclusion

How ECI is distributed across countries?

Southern Africa has the highest average ECI, reflecting a historical concentration of industrial and service-based economies.

Central Africa consistently shows the lowest ECI, with most countries losing complexity over time.

East Africa contains a mix of moderate and improving countries, including some of the region's best performers.

West Africa shows wide variation, from very low to moderate ECI, with no clear regional trend.

How ECI changed over time?

The region has seen a slow, modest improvement in economic complexity over the past two decades. The average country today is less negative than in the 1990s.

Former regional leaders (e.g., South Africa) have lost complexity, while a few smaller economies (e.g., Tanzania) have shown consistent improvement.

Which variables are related to ECI?

Strongest positive relationships: Tourism arrivals and CO₂ per capita – countries that are more visited and have higher emissions tend to have higher ECI. This suggests that industrial activity and global connectivity are the main drivers of complexity in the region.

Moderate positive relationships: GDP per capita and air transport – richer and better-connected countries are generally more complex, but the relationship is not linear.

Negative relationships: Energy use per unit of GDP (inefficiency) and coal rents – countries that rely on coal or use energy intensively tend to have lower ECI, indicating a resource curse and inefficiency penalty.

Do richer, cleaner, more visited countries have higher ECI?

Richer countries generally have higher ECI, but the relationship is not automatic – many middle-income countries perform as poorly as low-income ones.

Cleaner countries (lower CO₂ per capita) do not have higher ECI. In fact, the opposite is observed: higher emissions are associated with higher complexity, because industrial activity drives both.

More visited countries do have higher ECI. Tourism arrivals are one of the strongest positive correlates, indicating that global connectivity and openness support economic complexity.